

PCA-Conditioned QUBO Stress Testing for Long-Short Equity Portfolios

Project documentation generated from `unit.ipynb`

June 2, 2026

Abstract

This report documents the current state of the notebook `unit.ipynb`. The workflow builds a scenario-conditioned stress test for a unit-gross long-short equity portfolio. It uses exponentially weighted PCA to define systematic factor scenarios, constructs per-asset standardized-return grids around a selected factor scenario, converts those grids to simple-return states, and solves a sparse QUBO-style binary quadratic model using simulated annealing. The current implementation no longer builds volatility states; the state space is purely an asset-level return grid.

1 Project Overview

The model separates systematic movement from residual asset movement:

- exponentially weighted PCA defines a low-dimensional factor space;
- a Cartesian grid of factor scenarios defines systematic stress coordinates;
- a selected factor scenario is reconstructed into asset z-space;
- nearest residual donors are transplanted to the selected factor scenario and inflated when the scenario is far from the historical factor cloud;
- whole asset-level z bins are retained only when they lie inside an inflated residual range;
- z states are converted to log-return and simple-return states;
- a sparse compatibility QUBO penalizes implausible residual combinations while a portfolio term minimizes unit-gross PnL.

The notebook uses the first 100 assets from a hand-curated S&P-style ticker list. The historical period is January 1, 2015 through December 31, 2024. A 2025 backtesting period is declared but is not yet used.

2 Data and Standardization

Let $P_{t,i}$ denote the adjusted close of asset i on day t . The notebook uses daily log-returns

$$r_{t,i}^{\log} = \log \left(\frac{P_{t,i}}{P_{t-1,i}} \right). \quad (1)$$

Each asset is standardized independently:

$$z_{t,i} = \frac{r_{t,i}^{\log} - \mu_i}{\sigma_i}, \quad (2)$$

Table 1: Main settings and current run outputs.

Quantity	Current value
Number of assets	100
Historical observations after cleaning	933
Historical period	2015-01-01 to 2024-12-31
Declared backtest period	2025-01-01 to 2025-12-31
PCA factors retained	2
EW-PCA lookback window	125 days
EW-PCA decay	$\lambda = 0.93$
Factor-grid dimensions	21×5
Number of factor scenarios	105
Selected scenario index	5
Candidate z bins per asset	21
Kept z bins per asset in current run	17 to 21
Binary variables in current BQM	2,096
Quadratic interactions in current BQM	130,964
One-hot penalty	$\lambda_{\text{oh}} = 1$
Compatibility penalty	$\lambda_{\text{compat}} = 0.1$
Simulated annealing reads	100
Simulated annealing sweeps	5,000

where μ_i and σ_i are the historical mean and standard deviation of asset i 's log-return. In matrix notation,

$$Z \in \mathbb{R}^{T \times N}, \quad T = 933, \quad N = 100. \quad (3)$$

3 Exponentially Weighted PCA

The current notebook no longer fits ordinary equal-weight PCA. It fits PCA from the latest $L = 125$ standardized observations using exponential weights

$$\tilde{w}_t = \lambda^{a_t}, \quad \lambda = 0.93, \quad (4)$$

where $a_t = 0$ for the newest observation, $a_t = 1$ for the previous observation, and so on. The normalized weights are

$$w_t = \frac{\tilde{w}_t}{\sum_{u=1}^L \tilde{w}_u}. \quad (5)$$

The weighted mean is

$$\bar{z}^{\text{EW}} = \sum_{t=1}^L w_t z_t, \quad (6)$$

and the weighted covariance matrix is

$$C^{\text{EW}} = \sum_{t=1}^L w_t (z_t - \bar{z}^{\text{EW}})(z_t - \bar{z}^{\text{EW}})^\top. \quad (7)$$

The notebook eigendecomposes C^{EW} , sorts eigenvalues descending, and keeps $K = 2$ loading vectors. Let

$$W_K \in \mathbb{R}^{K \times N} \quad (8)$$

denote the retained loading matrix and let λ_k^{PCA} denote the retained eigenvalues. Factor scores for all historical observations are then computed as

$$f_t = (z_t - \bar{z}^{\text{EW}})W_K^\top. \quad (9)$$

The inverse reconstruction from factor space to standardized asset space is

$$\hat{z}(f) = \bar{z}^{\text{EW}} + fW_K. \quad (10)$$

The current run reports factor standard deviations

$$\sqrt{\lambda_1^{\text{PCA}}} = 5.0573, \quad \sqrt{\lambda_2^{\text{PCA}}} = 2.8865. \quad (11)$$

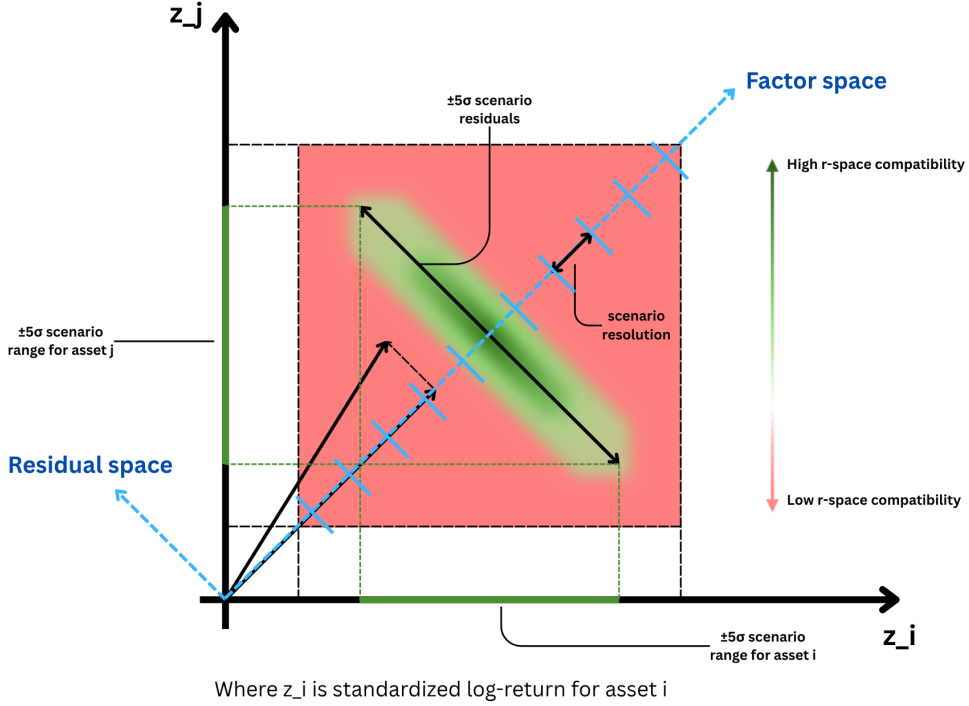


Figure 1: Conceptual relation between factor space, residual space, and asset-level standardized returns. The current implementation estimates the PCA basis from an exponentially weighted covariance matrix.

4 Factor Scenario Grid

The scenario grid is constructed in the two retained PC directions. For each PC, the grid uses integer multiples of the corresponding PCA eigen-sigma. If PC k has n_k grid points, define

$$u_{k,m} \in [-1, 1], \quad m = 1, \dots, n_k, \quad (12)$$

and

$$g_{k,m} = h_k \text{sign}(u_{k,m}) |u_{k,m}|^\gamma \sqrt{\lambda_k^{\text{PCA}}}, \quad h_k = \frac{n_k - 1}{2}. \quad (13)$$

The current notebook uses

$$(n_1, n_2) = (21, 5), \quad \gamma = 1. \quad (14)$$

Thus PC1 spans $[-10, +10]$ eigen-sigma units and PC2 spans $[-2, +2]$ eigen-sigma units. Their Cartesian product gives

$$21 \times 5 = 105 \quad (15)$$

factor scenarios.

The current selected scenario is index 5:

$$f_0 = \begin{bmatrix} -45.5160 \\ -5.7730 \end{bmatrix}. \quad (16)$$

In grid-bin terms this is approximately -9 PC1 bins and -2 PC2 bins from the factor origin.

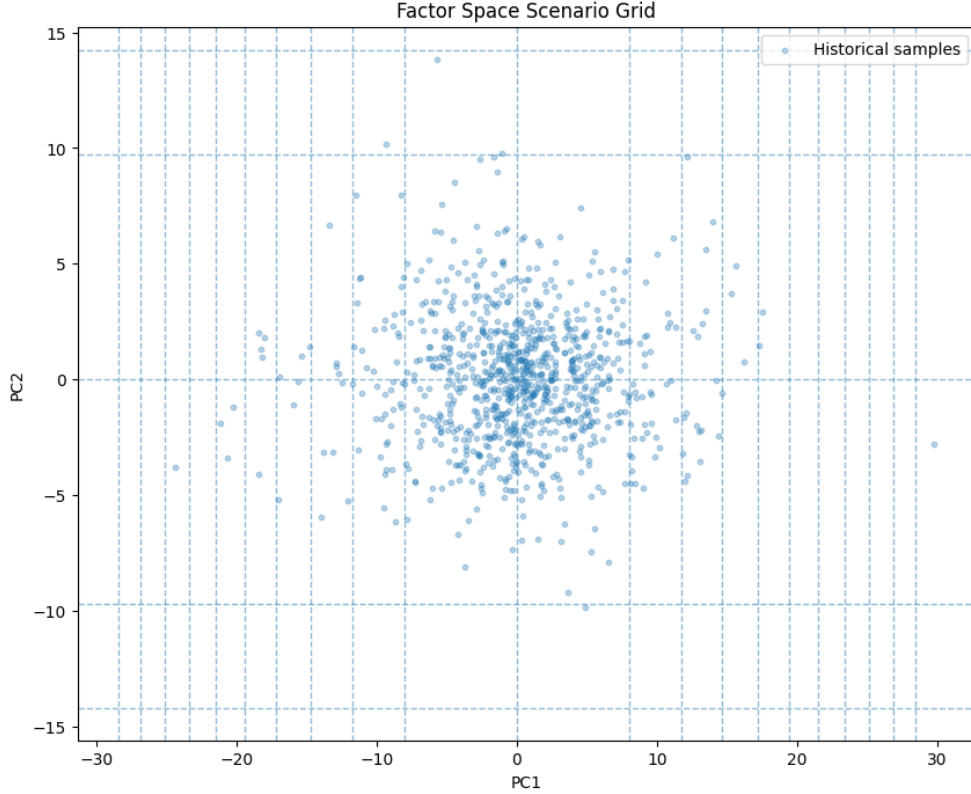


Figure 2: Factor-space scenario grid over historical factor samples. The current grid uses integer multiples of weighted PCA eigen-sigmas.

5 Scenario-Conditioned Asset Z Grids

For the selected factor scenario, the notebook first reconstructs a standardized asset-space scenario point:

$$\hat{z}(f_0) = \bar{z}^{\text{EW}} + f_0 W_K. \quad (17)$$

Historical PCA residuals are

$$\varepsilon_t = z_t - \hat{z}(f_t). \quad (18)$$

The notebook uses the most recent EW window for local conditioning and computes scaled factor-space distances

$$d_t = \left\| \frac{f_t - f_0}{(\sqrt{\lambda_1^{\text{PCA}}}, \dots, \sqrt{\lambda_K^{\text{PCA}}})} \right\|_2. \quad (19)$$

The 100 nearest recent factor observations are selected as residual donors. For the current run, the mean scaled distance of these donors from the selected scenario is

$$\bar{d} = 9.1066. \quad (20)$$

Because this scenario is far from the historical factor cloud, the donor residuals are inflated:

$$\alpha_{\text{infl}} = \min(1 + 0.50 \bar{d}^2, 5.0). \quad (21)$$

The current run hits the cap:

$$\alpha_{\text{infl}} = 5.0. \quad (22)$$

The inflated synthetic residual samples are

$$\varepsilon_t^{\text{infl}} = \alpha_{\text{infl}} \varepsilon_t, \quad (23)$$

and the corresponding synthetic scenario z samples are

$$z_t^{\text{synthetic}}(f_0) = \hat{z}(f_0) + \varepsilon_t^{\text{infl}}. \quad (24)$$

For asset i , the notebook also computes the single largest historical absolute standardized daily move

$$M_i = \max_t |z_{t,i}|. \quad (25)$$

This defines the full candidate range around the scenario reconstruction:

$$[\hat{z}_i(f_0) - M_i, \hat{z}_i(f_0) + M_i]. \quad (26)$$

That range is split into 21 equal candidate bins. Separately, an inflated residual range is formed from the local residual standard deviation s_i^ε :

$$[\hat{z}_i(f_0) - 5 \alpha_{\text{infl}} s_i^\varepsilon, \hat{z}_i(f_0) + 5 \alpha_{\text{infl}} s_i^\varepsilon]. \quad (27)$$

Only whole candidate bins whose left and right edges both lie inside the inflated residual range are retained. If a residual boundary cuts through a candidate bin, that bin is not included.

The resulting state count is asset-specific. The current run keeps between 17 and 21 z states per asset, for a total of 2,096 states across 100 assets. For NVDA, all 21 candidate z bins are retained:

$$\begin{aligned} &(-8.5751, -8.0853, -7.5955, -7.1057, -6.6159, -6.1261, -5.6363, \\ &-5.1465, -4.6567, -4.1669, -3.6771, -3.1873, -2.6975, -2.2077, \\ &-1.7179, -1.2281, -0.7383, -0.2485, 0.2413, 0.7311, 1.2209). \end{aligned} \quad (28)$$

6 Conversion from Z States to Return States

The current notebook creates `asset_grids` from `asset_z_grids`. For each asset i and z state s ,

$$r_{i,s}^{\text{log}} = \mu_i + \sigma_i z_{i,s}. \quad (29)$$

Because the underlying historical returns are log-returns, portfolio PnL is computed using simple returns

$$r_{i,s}^{\text{simple}} = \exp(r_{i,s}^{\text{log}}) - 1. \quad (30)$$

Each asset grid stores three aligned arrays:

$$\text{asset_grids}[i] = \{z, r^{\text{log}}, r^{\text{simple}}\}. \quad (31)$$

For the current NVDA grid, the simple-return states range from approximately -13.0870% to $+2.0983\%$.

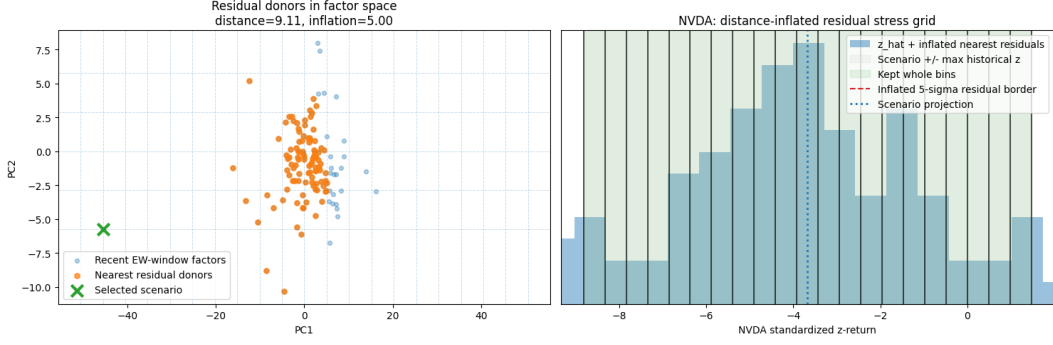


Figure 3: Illustrative local z-grid visualization for NVDA. The current notebook visualization marks full candidate bin edges, retained whole bins, the inflated residual boundary, and the scenario projection.

7 Portfolio Model

The current notebook builds a synthetic correlation-hedged long-short portfolio rather than simple independent random weights. Five anchor positions are drawn, assigned 60% of gross exposure, and then hedged using correlated non-anchor names. The hedge basket receives the remaining 40% of gross exposure, and a small 2% gross idiosyncratic noise basket is added. The final portfolio is net-corrected and gross-normalized:

$$\sum_{i=1}^N |w_i| = 1, \quad \sum_{i=1}^N w_i \approx 0. \quad (32)$$

The current run reports exactly unit gross exposure and zero net exposure to printed precision.

Table 2: Largest current portfolio positions by absolute weight.

Ticker	Weight
KLAC	-0.27135
BX	-0.20131
BLK	0.06055
RTX	-0.01682
NEM	0.01319
ADI	-0.01051
INTC	0.00866
JPM	0.00864
CSCO	0.00764
C	0.00755

For any selected state vector, the reported PnL is

$$\Pi = \sum_{i=1}^N w_i r_{i,s_i}^{\text{simple}}. \quad (33)$$

Since $\sum_i |w_i| = 1$, Π is a fractional PnL per unit gross exposure.

8 Binary Quadratic Model

For every asset i and retained state s , define

$$x_{i,s} \in \{0, 1\}. \quad (34)$$

The interpretation is

$$x_{i,s} = 1 \iff \text{asset } i \text{ selects state } s. \quad (35)$$

State counts are no longer uniform. Let S_i denote the number of retained states for asset i . The current run has

$$17 \leq S_i \leq 21, \quad \sum_{i=1}^{100} S_i = 2,096. \quad (36)$$

8.1 One-Hot Constraint

Each asset must choose exactly one state:

$$E_{\text{oh}}(x) = \lambda_{\text{oh}} \sum_{i=1}^N \left(\sum_{s=1}^{S_i} x_{i,s} - 1 \right)^2. \quad (37)$$

The current penalty is

$$\lambda_{\text{oh}} = 1. \quad (38)$$

For binary variables,

$$\left(\sum_s x_{i,s} - 1 \right)^2 = - \sum_s x_{i,s} + 2 \sum_{s < t} x_{i,s} x_{i,t} + 1. \quad (39)$$

Thus zero selected states or two selected states both create a one-unit penalty for that asset; three selected states create a four-unit penalty.

8.2 Rejected Dense Factor Penalty

The notebook intentionally does not use a dense global factor penalty of the form

$$E_{\text{factor}}(x) = \lambda_f \|W_K z(x) - f_0\|_2^2. \quad (40)$$

Such a term would align the decoded state more directly with f_0 , but it would couple nearly every asset-state variable to nearly every other asset-state variable. The current implementation relies instead on scenario-conditioned asset grids plus sparse residual compatibility.

8.3 Sparse Residual Compatibility

For a state s of asset i , define the implied residual relative to the scenario reconstruction:

$$\varepsilon_{i,s} = z_{i,s} - \hat{z}_i(f_0). \quad (41)$$

The compatibility residual sample is the inflated residual donor matrix when it is available. From it, the notebook estimates residual covariance and correlation matrices. For each asset, the five largest absolute correlations define a sparse neighbor set; duplicate directed edges are skipped.

For a neighboring pair (i, j) with residual correlation ρ_{ij} and residual standard deviations σ_i^ε and σ_j^ε , the Gaussian conditional expectation is

$$\mathbb{E}[\varepsilon_j \mid \varepsilon_i] = \rho_{ij} \frac{\sigma_j^\varepsilon}{\sigma_i^\varepsilon} \varepsilon_i, \quad (42)$$

with conditional variance

$$\text{Var}(\varepsilon_j \mid \varepsilon_i) = (\sigma_j^\varepsilon)^2 (1 - \rho_{ij}^2). \quad (43)$$

The pairwise compatibility penalty is

$$C_{ij}(s, t) = \frac{\left(\varepsilon_{j,t} - \rho_{ij} \frac{\sigma_j^\varepsilon}{\sigma_i^\varepsilon} \varepsilon_{i,s}\right)^2}{(\sigma_j^\varepsilon)^2 (1 - \rho_{ij}^2)}. \quad (44)$$

The structural compatibility energy is

$$E_{\text{compat}}(x) = \lambda_{\text{compat}} \sum_{(i,j) \in \mathcal{G}} \sum_{s=1}^{S_i} \sum_{t=1}^{S_j} C_{ij}(s, t) x_{i,s} x_{j,t}, \quad (45)$$

with current scale

$$\lambda_{\text{compat}} = 0.1. \quad (46)$$

The raw $C_{ij}(s, t)$ term is dimensionless: it is a squared residual mismatch measured in conditional-sigma units.

8.4 Portfolio Objective

The portfolio objective uses simple-return states:

$$E_{\text{port}}(x) = \sum_{i=1}^N \sum_{s=1}^{S_i} w_i r_{i,s}^{\text{simple}} x_{i,s}. \quad (47)$$

Minimizing this term searches for low portfolio PnL. The total energy is

$$E(x) = E_{\text{oh}}(x) + E_{\text{compat}}(x) + E_{\text{port}}(x). \quad (48)$$

8.5 Energy Units and Scaling

Because the portfolio is normalized to unit gross exposure and the portfolio term uses simple returns, E_{port} is measured in fractional PnL per unit gross exposure. A value of -0.01 means a loss of 1% of gross exposure.

The one-hot and compatibility terms are artificial QUBO penalties added to the same scalar energy. They are not portfolio PnL, but their numerical scales compete with the portfolio term during optimization. With $\lambda_{\text{oh}} = 1$, a broken one-hot constraint costs one full QUBO energy unit, which is typically much larger than a single portfolio PnL term. With $\lambda_{\text{compat}} = 0.1$, a one-conditional-sigma residual mismatch on one compatibility edge costs approximately 0.1 energy units.

9 Optimization and Current Diagnostics

The current total BQM has

$$2,096 \quad (49)$$

binary variables and

$$130,964 \quad (50)$$

quadratic interactions. It is solved using simulated annealing with 100 reads and 5,000 sweeps. The current best reported energy is

$$E_{\text{min}} = 5.253112. \quad (51)$$

After decoding the best sample, the selected standardized asset state is projected back into the EW-PCA factor space:

$$\hat{f}_{\text{selected}} = (z_{\text{selected}} - \bar{z}^{\text{EW}}) W_K^\top. \quad (52)$$

The current decoded factor state is

$$\hat{f}_{\text{selected}} = \begin{bmatrix} -46.1901 \\ -9.4609 \end{bmatrix}, \quad (53)$$

compared with the target

$$f_0 = \begin{bmatrix} -45.5160 \\ -5.7730 \end{bmatrix}. \quad (54)$$

The Euclidean factor reconstruction error is

$$\|\hat{f}_{\text{selected}} - f_0\|_2 \approx 3.75. \quad (55)$$

The notebook also reports error in scenario-grid bin units:

$$\Delta_{\text{PC1}} = -0.13 \text{ bins}, \quad \Delta_{\text{PC2}} = -1.28 \text{ bins}. \quad (56)$$

Thus the selected state is close to the target in PC1 grid units but more than one PC2 bin below the requested scenario. This is expected behavior for the current sparse model because no dense factor-matching penalty is imposed.

The reported portfolio PnL is

$$\Pi_{\text{selected}} = -0.8144\% \quad (57)$$

of gross portfolio value. Since the portfolio is unit-gross normalized, this is also the QUBO portfolio objective scale expressed as a percentage.

10 Scientific Interpretation

The current model is a constrained discrete search over scenario-conditioned asset return states. The conditioning operation places each asset’s candidate states near $\hat{z}(f_0)$, while residual transplanting and inflation provide a pragmatic way to populate severe extrapolative factor scenarios. The sparse compatibility graph approximates pairwise residual dependence:

$$p(\varepsilon \mid f = f_0) \approx \prod_{(i,j) \in \mathcal{G}} p(\varepsilon_j \mid \varepsilon_i). \quad (58)$$

This is not a normalized probability model. It is an energy model with three roles:

- one-hot terms enforce valid state selection;
- compatibility terms discourage conditionally implausible residual combinations;
- the portfolio term pushes the selected state toward adverse PnL.

The current factor-bin diagnostic is important. It shows whether sparse residual compatibility and local grids preserve the intended factor scenario well enough after the optimizer chooses portfolio-adverse states. In the current run, PC1 remains close to the target grid coordinate, while PC2 drifts by about 1.28 bins.

11 Limitations and Next Steps

The current notebook remains a prototype:

- The declared 2025 backtesting period is not yet used.
- The portfolio is synthetic, although it is now correlation-hedged and unit-gross normalized.

- The state space no longer contains volatility states; volatility risk is not modeled in the current QUBO.
- The residual transplant and distance-inflation scheme is pragmatic and should be calibrated before production use.
- Sparse residual compatibility does not guarantee exact factor matching; factor-bin diagnostics should be monitored scenario by scenario.
- Penalty scales are still heuristic. They should be swept relative to the independent one-hot worst-PnL scale.

Natural extensions include out-of-sample validation on the declared 2025 period, penalty-scaling sweeps, benchmark comparisons against smaller exact solves, realistic holdings, and optional reintroduction of volatility-sensitive payoff or constraint terms.

12 Conclusion

The current notebook implements an EW-PCA-conditioned QUBO stress-test prototype. Compared with earlier versions, it now uses exponentially weighted PCA, a PC-eigen-sigma scenario grid, distance-inflated residual donor samples, whole-bin z-state filtering, simple-return asset grids, a correlation-hedged unit-gross portfolio, and a smaller sparse QUBO with 2,096 variables in the current run. The reported decoded state produces a portfolio loss of 0.8144% of gross exposure and lands -0.13 PC1 bins and -1.28 PC2 bins from the selected factor scenario.